HR Insights Analysis Dashboard

By Eng/ Omar Ibrahim Elashry

Project Title: HR Insights Analysis Dashboard

Project Developer

Omar Ibrahim Elashry (Data Analyst & Business Development Manager)

LinkedIn: https://www.linkedin.com/in/omar-ibrahimelashry

Email: omarelashry27@gmail.com

Project Overview

This project presents a comprehensive human resources analytics dashboard designed to monitor key workforce metrics, including employee count, attrition rates, demographic distributions, job role performance, and compensation trends. The dashboard supports strategic HR decision-making by identifying risk areas and patterns in employee turnover.

Tools Used

- Database from (Kaggle)
- Data visualization tool (Microsoft Power BI)

- Data aggregation and cleaning (Microsoft Excel, Python Programming Language)

Key Features & Insights

1. Workforce Overview

- Total Employees: 1,416
- Overall Attrition Rate: 16.2%
- Highlights a significant turnover level requiring further investigation and retention strategies.

2. Attrition Analysis by Gender

- Male attrition: 136
- Female attrition: 76
- Indicates higher turnover among male employees, suggesting potential gender-specific workplace factors.

3. Attrition by Job Role

- Highest attrition observed in Laboratory Technicians (60) and Sales Representatives (55).
- Roles like Research Director (2) and Manager (5) show very low turnover.
- Useful for targeting retention efforts toward high-attrition roles.

4. Education Demographics across Departments

- Visualizes employee distribution by education level and department.
- Largest department: Medical (229).
- Supports workforce planning and hiring alignment with educational backgrounds.

5. Attrition Analysis by Salary Slab

- Up to 5k: 158 employees left
- 5k–10k: 48
- 10k–15k: 18
- 15k+: 5

6. Average Working Years by Job Role

- Compares tenure across roles such as Manager, Researcher, Healthcare, and Sales Executive.
- Helps identify roles with shorter average tenure, which may indicate burnout or lack of growth.

7. Attrition Trends across Education Fields

- Medical (25%) and Marketing (15%) account for the largest share of attrition.
- Other (5%) shows the lowest.
- Enables targeted retention programs for high-attrition education groups.

Business Value

- Provides clear visibility into attrition drivers by gender, role, salary, and education
- Supports retention strategy development and workforce planning
- Helps identify high-risk employee segments for proactive intervention
- Enables data-driven HR policies to reduce turnover costs